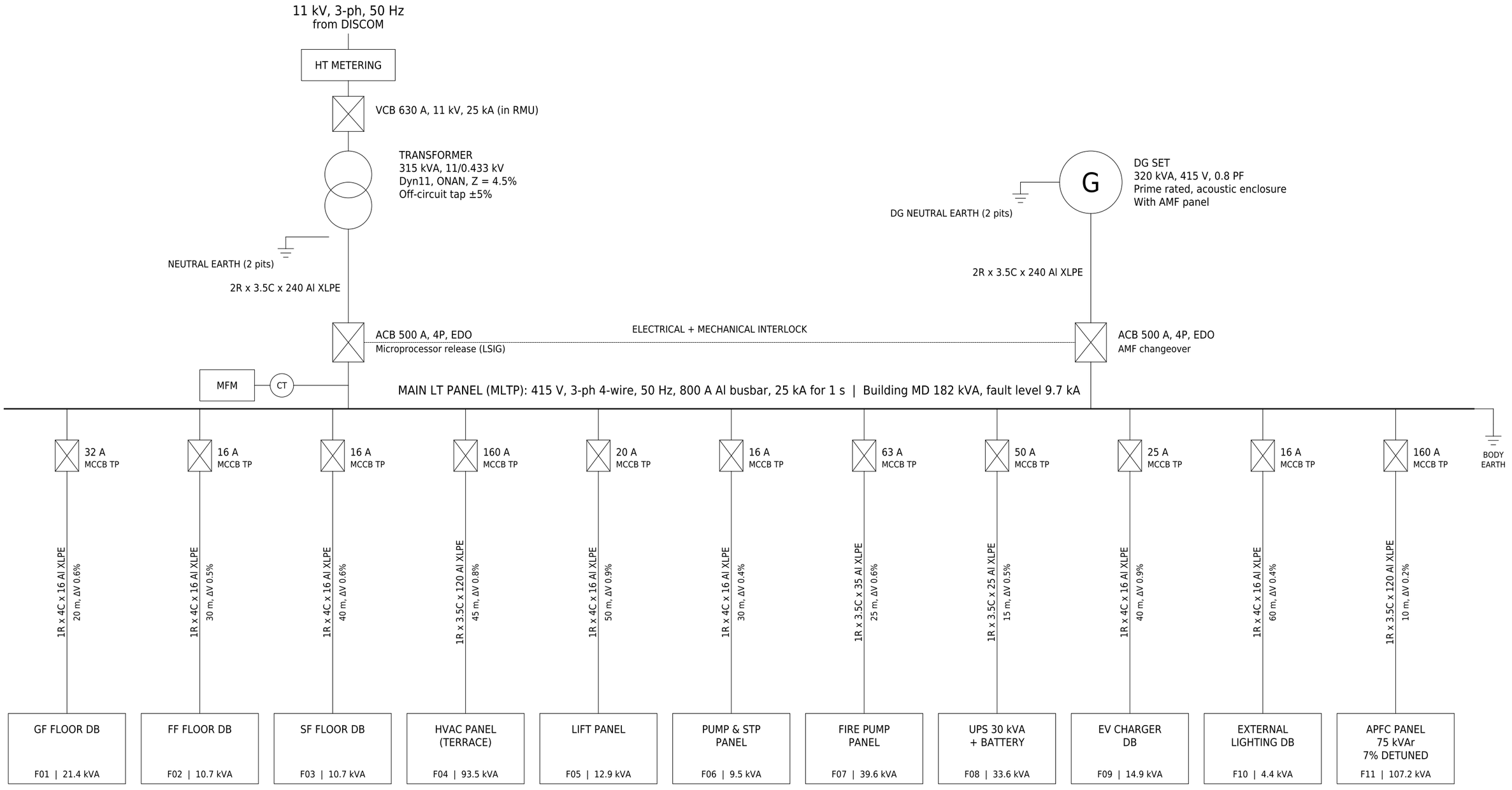
SINGLE LINE DIAGRAM: G+2 COMMERCIAL OFFICE BUILDING, 11 kV / 415 V



F07: fire-survival cable, separate route

NOTES

1. All ratings derived in G2_Office_Electrical_Design_Calcs.xlsx (Load_Schedule, Sizing, Cable_Schedule).
2. LT cables: Havells 0.6/1 kV Al XLPE armoured, 3.5 core (4 core for 16 mm²), on tray; derating 0.87 (45°C) x 0.71 (grouping).
3. Feeder voltage drop ≤ 3% (IS 732 / IEC 60364-5-52); end-to-end ≤ 5%.
4. Earthing: TN-S. Main grid 7 pipe electrodes + 50 x 6 GI ring, calc. R = 0.99 Ω (IS 3043). Main earth conductor 32 x 6 GI.
5. Fire pump feeder taken ahead of DG changeover per NBC 2016 Part 4 (shown from MLTP for clarity).
6. Breaking capacity of MCCBs ≥ 25 kA. Discrimination to be verified with manufacturer curves.

PROJECT	G+2 Commercial Office, Delhi
DRAWING	Single Line Diagram, LT & HT
DRAWN BY	Anuj Kumar
SCALE	NTS SHEET: A3
DWG NO.	G2-EL-SLD-001 REV 0